



Main power transformers for solar and storage projects.

Standard configurations, 30–100 MVA at 115 and 138 kV.

Engineered to IEEE C57, factory-tested with third-party witness,
delivered to your site on a committed date.

Built to spec. Delivered on a date.

Farpoint Energy supplies main power transformers to solar and storage developers who cannot wait three years for a substation to be energized. We design to a small set of standard configurations, manufacture them at Farpoint-qualified partner facilities under our own quality program, and stand behind every unit as the supplier of record: our contract, our factory acceptance test, our warranty, our service response.

Fixed configurations are the reason we can commit to a delivery month. A repeat design means drawings exist on day one, long-lead materials are reserved before your order is signed, and the factory has built the unit before. Custom is what the 2–4 year market sells. Standard is how we deliver in 12–14 months.

Transformers are where Farpoint starts. Our purpose is to build the equipment the grid is waiting for — and, over time, to make abundant power cheaper and faster to deploy than it has ever been.

12–14

MONTHS TO SITE

30–100

MVA, THREE-STAGE
COOLING

115 / 138

KV HIGH SIDE

C57

IEEE DESIGN &
TEST

Supplier of record. The purchase contract, the letter of credit to the factory, the factory acceptance test, importer-of-record duties, and the warranty all sit with Farpoint Energy — a US entity. You buy a transformer from us; how we build it is our responsibility.

How Farpoint delivers

01 Configuration match

Send your single-line diagram and site data. We confirm the standard configuration that fits, and any project-specific values (impedance, LV connection, BIL) within the design window.

02 Firm quote, committed month

Price, delivery month, and terms — binding — within ten business days.

03 Drawings for approval

Outline, nameplate, and test plan issued to your engineer and interconnecting utility for review. The approval cycle is built into the schedule.

04 Build and witnessed test

Production in a reserved slot. Full routine test suite to IEEE C57.12.90, witnessed by an independent inspection agency. You are welcome at FAT.

05 Delivery and energization

Export packing with impact recorders, ocean and inland transport, DAP to your site. Commissioning supervision at energization. US-based service response under warranty.

Standard configurations

Every unit: 60 Hz · ONAN/ONAF/ONAF three-stage cooling · 65 °C average winding rise · 34.5 kV delta low side (wye per project) · DETC $\pm 2 \times 2.5$ % on the high side · mineral oil · designed and tested to ANSI/IEEE C57.12.00 and C57.12.90.

CONFIGURATION	RATING (MVA)	HIGH SIDE	LOW SIDE	TYPICAL PROJECT FIT	DATASHEET
FP-MPT-138-50	30 / 40 / 50	138 kV · 650 kV BIL	34.5 kV · 200 kV BIL	50–60 MW block, one unit	p. 4
FP-MPT-115-50	30 / 40 / 50	115 kV · 550 kV BIL	34.5 kV · 200 kV BIL	50–60 MW block, one unit	p. 4
FP-MPT-138-100	60 / 80 / 100	138 kV · 650 kV BIL	34.5 kV · 200 kV BIL	80–110 MW block, one unit	p. 5
FP-MPT-115-100	60 / 80 / 100	115 kV · 550 kV BIL	34.5 kV · 200 kV BIL	80–110 MW block, one unit	p. 5
Two-unit configurations	2 × 50 or 2 × 100 class	115 / 138 kV	34.5 kV	120–200 MW; phased energization and N-1 redundancy	on request
161 kV variant	per project	161 kV	34.5 kV	Quoted on request — different insulation class	on request

Sizing guide

Total transformer MVA $\approx$ project MW $\div$ 0.95, one unit per 100–150 MW block. A 55 MW project at 138 kV is one FP-MPT-138-50. A 95 MW project is one FP-MPT-138-100. A 160 MW project is two units, which also lets you energize the first block while the second is still under construction.

Send us the single-line diagram; we confirm the fit and flag anything in your spec that would push the unit outside the standard window.

Why a rating reads 30 / 40 / 50

One transformer, three cooling stages. 30 MVA with oil and air circulating naturally (ONAN); 40 MVA with the first bank of fans running (ONAF); 50 MVA with the second bank (ONAF). Same core, same windings. You run quiet at base load and let the fans carry the afternoon peak — the staged rating is matched to a solar generation profile, and it is the standard IEEE way to rate a unit of this class.

FP-MPT-138-50 · FP-MPT-115-50

30 / 40 / 50 MVA main power transformer

Rev. A · 2026

PARAMETER	FP-MPT-138-50	FP-MPT-115-50
RATINGS		
Rated power	30 / 40 / 50 MVA (ONAN / ONAF / ONAF)	30 / 40 / 50 MVA (ONAN / ONAF / ONAF)
Frequency	60 Hz	60 Hz
Temperature rise	65 °C average winding, 80 °C hottest spot	65 °C average winding, 80 °C hottest spot
Phases / vector group	3-phase, YNd1 (typical; per project)	3-phase, YNd1 (typical; per project)
VOLTAGE & INSULATION		
High-voltage winding	138 kV, grounded wye – 145 kV class	115 kV, grounded wye – 123 kV class
HV basic impulse level	650 kV (550 kV with surge arresters, per study)	550 kV (450 kV with surge arresters, per study)
Low-voltage winding	34.5 kV, delta (wye available per single-line)	34.5 kV, delta (wye available per single-line)
LV basic impulse level	200 kV	200 kV
Impedance	8.5 % at 50 MVA base (7.5–9 % window, per project)	8.5 % at 50 MVA base (7.5–9 % window, per project)
Taps	DETC on HV, $\pm 2 \times 2.5$ %, no-load · LTC optional	DETC on HV, $\pm 2 \times 2.5$ %, no-load · LTC optional
CONSTRUCTION		
Core / windings	Grain-oriented electrical steel core; copper windings	Grain-oriented electrical steel core; copper windings
Insulating fluid	Mineral oil (natural-ester FR3-class optional)	Mineral oil (natural-ester FR3-class optional)
Oil preservation	Sealed tank with nitrogen blanket, or conservator	Sealed tank with nitrogen blanket, or conservator
Cooling	Detachable radiators with two fan banks; fan control by winding temperature	Detachable radiators with two fan banks; fan control by winding temperature
STANDARDS & SITE		
Design and test	ANSI/IEEE C57.12.00 (general), C57.12.90 (test code), C57.91 (loading guide), NEMA TR-1 (sound)	
Site basis	Outdoor, altitude $\leq 1,000$ m, 30 °C average / 40 °C maximum ambient; seismic per site	

Sound level per NEMA TR-1 · ANSI 70 light gray finish · standard accessories
listed on page 6 · outline drawings and weights issued with the quotation.

FP-MPT-138-100 · FP-MPT-115-100

60 / 80 / 100 MVA main power transformer

Rev. A · 2026

PARAMETER	FP-MPT-138-100	FP-MPT-115-100
RATINGS		
Rated power	60 / 80 / 100 MVA (ONAN / ONAF / ONAF)	60 / 80 / 100 MVA (ONAN / ONAF / ONAF)
Frequency	60 Hz	60 Hz
Temperature rise	65 °C average winding, 80 °C hottest spot	65 °C average winding, 80 °C hottest spot
Phases / vector group	3-phase, YNd1 (typical; per project)	3-phase, YNd1 (typical; per project)
VOLTAGE & INSULATION		
High-voltage winding	138 kV, grounded wye – 145 kV class	115 kV, grounded wye – 123 kV class
HV basic impulse level	650 kV (550 kV with surge arresters, per study)	550 kV (450 kV with surge arresters, per study)
Low-voltage winding	34.5 kV, delta (wye available per single-line)	34.5 kV, delta (wye available per single-line)
LV basic impulse level	200 kV	200 kV
Impedance	Per project; confirmed at quotation	Per project; confirmed at quotation
Taps	DETC on HV, $\pm 2 \times 2.5 \%$, no-load · LTC optional	DETC on HV, $\pm 2 \times 2.5 \%$, no-load · LTC optional
CONSTRUCTION		
Core / windings	Grain-oriented electrical steel core; copper windings	Grain-oriented electrical steel core; copper windings
Insulating fluid	Mineral oil (natural-ester FR3-class optional)	Mineral oil (natural-ester FR3-class optional)
Oil preservation	Conservator with rubber bag, or sealed tank with nitrogen blanket	Conservator with rubber bag, or sealed tank with nitrogen blanket
Cooling	Detachable radiators with two fan banks; fan control by winding temperature	Detachable radiators with two fan banks; fan control by winding temperature
STANDARDS & SITE		
Design and test	ANSI/IEEE C57.12.00 (general), C57.12.90 (test code), C57.91 (loading guide), NEMA TR-1 (sound)	
Site basis	Outdoor, altitude $\leq 1,000$ m, 30 °C average / 40 °C maximum ambient; seismic per site	

Sound level per NEMA TR-1 · ANSI 70 light gray finish · standard accessories listed on page 6 · outline drawings and weights issued with the quotation.

OPTIONS

Configure within the standard

Options are pre-engineered so they do not reopen the design. Each is quoted as a line item with its effect on delivery stated plainly.

OPTION	WHAT IT IS	WHEN YOU'D SPECIFY IT
On-load tap changer	LTC in place of DETC, ± 16 steps typical; current component lead time stated in the quote	Utility requires voltage regulation under load
Natural-ester fluid	FR3-class biodegradable, high fire-point fluid in place of mineral oil	Fire-safety setbacks, environmental permits
Reduced BIL	550 kV (138 kV) or 450 kV (115 kV) with surge arresters	Where your insulation coordination study supports it
Harmonic-duty review	Design check against your inverter and storage duty cycle; derating or design adjustment if required	Storage-heavy sites; high inverter harmonic content
Monitoring package	Online dissolved-gas analysis, bushing monitoring, digital temperature and cooling control with SCADA points	Owners running condition-based maintenance
Spare-unit program	A reserved production slot or a stocked spare held against your operating fleet	Fleets that cannot tolerate a multi-year replacement wait
Extended warranty	Beyond the standard term, with defined service response	Lender or owner requirements

Standard accessories (all configurations)

ANSI outdoor porcelain bushings · multi-ratio bushing current transformers (relay classes per project) · sudden-pressure relay · top-oil and winding temperature indicators with alarm/trip contacts · liquid-level gauge · pressure-relief device · dissolved-gas monitoring provision · drain and sampling valves · jacking pads and skid base with provision for rail or trailer transport · grounding pads · nameplate per C57.12.00.

QUALITY

Every unit is proven, not assumed

Where it is built

Farpoint-qualified manufacturing partners operating ISO 9001 quality systems with accredited high-voltage test laboratories. Each partner is qualified against our design standard, process audit, and in-process hold points before it builds a Farpoint unit.

Design authority

Drawings are issued and controlled by Farpoint. Your engineer and interconnecting utility review outline drawings, nameplate, and the test plan before manufacturing releases.

Independent inspection

An independent inspection agency (SGS or Bureau Veritas class) witnesses in-process hold points, the factory acceptance test, and pre-shipment packing on every unit. Customers and owner's engineers are welcome at FAT.

Factory acceptance tests — IEEE C57.12.90

- Winding resistance · ratio, polarity and phase relation
- No-load losses and excitation current
- Load losses and impedance at rated current
- Applied-voltage and induced-voltage tests, with partial-discharge measurement
- Lightning impulse test, full and chopped wave
- Insulation power factor · core ground · leak test
- Oil dielectric strength, moisture, and dissolved-gas baseline
- Temperature-rise test on the first unit of each configuration
- Sound level, when specified

Test reports, oil certificates, and the inspection agency's release are delivered with the unit and archived with its serial number.

DELIVERY

12–14 months, with the month in writing

The delivery month is in the quotation and in the purchase order. It covers the approval cycle, production, testing, ocean transport, and inland delivery to your site.

STAGE	WHAT HAPPENS
Order	Purchase order and deposit; production slot confirmed
Weeks 1–8	Drawings for approval; long-lead materials committed
Production	Core, windings, assembly, vacuum drying, oil processing
FAT	Full routine suite, witnessed; test report issued
Transit	Export packing with impact recorders; ocean and inland freight; Farpoint is importer of record
Site	DAP delivery; commissioning supervision at energization

What's included with every unit

- Complete drawing package for utility and owner's-engineer review
- Witnessed factory acceptance test per IEEE C57.12.90
- Temperature-rise test on the first unit of a configuration
- Export packing, ocean and inland transport, customs — DAP your site
- Commissioning supervision at energization
- Warranty: 18 months from energization or 24 months from shipment
- US-based field service response under warranty
- Escalation-indexed pricing on published indices, the same structure utility RFPs use

Warranty and service. Warranty claims are handled by Farpoint, not passed to a factory abroad. Field response — oil handling, testing, diagnostics, minor repair — is provided by our US service partner; major repair or replacement is backed by the manufacturing partner under our contract.

HOW TO ORDER

Three steps to a committed month

01 Send the single-line diagram

Plus site data and your required delivery window. A short spec sheet is fine; we translate it into the configuration and flag any deviation.

02 Receive a firm quote within ten business days

Price, delivery month, terms, and the inspection regime — binding for a stated validity period.

03 Order

Purchase order with deposit; milestone payments at drawing approval, factory acceptance test, and shipment.



FARPOINT ENERGY

Request a configuration match or a quote.

Send a single-line diagram and a delivery window. We reply with the configuration that fits and a firm quote within ten business days.

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